

Machine / Line Downtime Recording and Escalation

Operations Agent Knowledge Base • Draft template for plant-specific review

Document ID	Version	Process owner	Effective date
SOP-OPS-003	1.0 (Draft)	Production / Operations (confirm locally)	After approval

Important document status

This is a draft template, not an approved plant instruction. Before operational use, confirm the plant's downtime definitions, MES fields and event codes, role ownership, escalation contacts, communication channels, response-time targets, safety procedures, and approval requirements. Do not infer missing site rules or thresholds.

1. Purpose

Provide a consistent method to detect, record, classify, review, and escalate machine or production-line downtime. The Operations Agent supports record review and summarization; it does not replace operator judgment, maintenance procedures, or safety controls.

2. Scope

Applies to downtime events represented in the site's MES or approved operations records, including planned stops, unplanned stops, changeovers, material shortages, quality-related stops, equipment faults, and events with an unknown cause. Use only categories configured and approved by the plant.

3. Roles and responsibilities

Role	Responsibility
Operator / line team	Follow local safety procedures; report the stop and provide accurate event details; record restart only as permitted by site procedure.
Shift supervisor	Review event completeness and classification, coordinate escalation using approved rules, and confirm shift-handover notes.
Maintenance / technical support	Investigate equipment-related events and record findings and actions in the approved system.
Production / operations manager	Review significant or recurring downtime, assign follow-up actions, and approve local workflow changes.
Operations Agent	Retrieve records, flag missing/conflicting fields, summarize events, and cite sources. Never fabricate causes or status.

4. Required downtime record fields

Field	Recording guidance
Event identifier	Use the unique MES event ID or approved reference. Do not invent an ID.
Machine / line	Use the plant's canonical asset or line identifier.
Start timestamp	Use the source timestamp and time zone; preserve source precision.
End / restart timestamp	Record only when confirmed. Mark an event ongoing only when supported by the source.
Duration	Prefer the MES-calculated duration. Otherwise calculate only from valid start/end timestamps and label it as calculated.
Downtime category / code	Use approved site codes. If unavailable, mark unclassified/not recorded according to local configuration.
Symptom and cause	Separate observed facts, suspected causes, and confirmed causes.
Impact	Include affected output, quality, or schedule only when supported by source data.
Action, owner, status	Use approved event/work-order records; do not infer an owner or completion status.

Field	Recording guidance
Evidence / references	Link the source event, work order, maintenance record, or approved note.

5. Procedure

5.1 Detect and confirm

Use the approved MES downtime event, line status, or operator report. Determine planned/unplanned status only using site definitions. Preserve and flag disagreements between sources.

5.2 Create or update the record

Search for an existing event before creating a new one to reduce duplicates. Enter required fields in the designated system.

5.3 Validate timestamps and duration

Check for missing timestamps, inconsistent time zones, end-before-start, negative duration, overlapping events, and conflicts. Flag anomalies instead of silently correcting them.

5.4 Classify the event

Select an approved category supported by evidence. Keep symptom, suspected cause, confirmed cause, and corrective action distinct.

5.5 Notify and escalate

Follow the approved site escalation matrix. Safety hazards must be handled immediately through the established emergency/safety process; never wait for agent analysis.

5.6 Investigate and document

The responsible team records findings, work-order references, corrective actions, owner, and status in the approved system. The agent may summarize but must not invent a diagnosis.

5.7 Confirm restart and closure

Record restart or closure only when confirmed by an authorized source or role. Keep unresolved events open/pending according to local workflow.

5.8 Shift handover

Include ongoing event, start time, known symptom, confirmed actions, pending owner/next step if recorded, and source reference. Mark unknown information clearly.

5.9 Review recurring events

Use approved reporting windows and categories to identify repeat stops or high cumulative duration. Patterns are prompts for investigation, not proof of root cause.

6. Escalation rules

The following is a configuration checklist, not a substitute for local response-time commitments or emergency procedures.

Condition	Required handling
Safety hazard, injury, or unsafe condition	Follow the established site emergency and safety reporting process immediately.
Production stop requiring technical support	Notify the designated supervisor/maintenance contact according to the approved matrix.
Missing, conflicting, or implausible event data	Flag for human review; retain source values and explain the discrepancy.
Long-duration or repeated downtime	Apply only site-defined duration/frequency thresholds. If none are configured, do not invent one; refer to the supervisor.
Cause unknown or action unresolved	Keep cause/status unknown or pending; identify an owner only if recorded.

7. Operations Agent behavior and guardrails

- Cite the source system, event/record ID, timestamp, and relevant field or document section where available.
- Distinguish observed facts, recorded causes, suspected causes, and agent-generated hypotheses.
- Do not create or alter MES records, close work orders, change machine settings, or issue control commands unless a separately approved integration explicitly authorizes that action.

- Never infer that a machine is safe to restart; follow local safety and restart procedures.
- Do not fabricate thresholds, event codes, contacts, duration, production loss, ownership, or resolution status.
- If evidence is missing or contradictory, say: **“Unable to verify from available records”** and explain what needs review.
- Restrict output to the user's authorized plant, line, asset, and record scope.

8. Suggested downtime summary format

Item	Expected content
Reporting window	Start/end date-time and time zone.
Scope	Plant, line, machine, and shift.
Downtime events	Count of source-backed events and their IDs/links.
Total downtime	Value and source/calculation method; identify incomplete events separately.
Categories	Counts/duration by approved code, with source references.
Ongoing events	Machine, start time, known symptom, status, and recorded owner/next action.
Data-quality notes	Missing timestamps, duplicates, overlaps, conflicts, and unclassified events.
Follow-up	Open actions, responsible role, and due date only where recorded.

9. Site-specific approval checklist

- Confirm scope, official downtime definitions, and planned/unplanned rules.
- Map required fields to actual MES tables, columns, event codes, and timestamp conventions.
- Approve duration calculation and duplicate-event handling rules.
- Add roles, escalation contacts, channels, and response-time targets.
- Define thresholds for prolonged/repeated downtime and the approving owner.
- Link the approved safety and restart procedures.
- Validate access control, audit trail, retention, and privacy requirements.
- Test: ongoing event, missing end time, duplicate, conflicting cause, invalid duration, and recurring downtime.
- Obtain process-owner and document-control approval before operational release.

10. Revision and approval record

Version	Date	Change summary	Prepared / reviewed / approved by
1.0 Draft	2026-09-30	Initial template for Operations Agent knowledge base.	Complete locally

End of SOP-OPS-003 • Draft template • Use only after site-specific validation and approval.